EXPERIMENT 17

AIM:

To implement a Linear Regression model for predicting marks based on study hours.

CODE:

```
import numpy as np
import pandas as pd

# Load the dataset
df = pd.read_csv('StudyHours_Marks.csv')
print(df)

# Display dataset information
df.info()

# Remove rows with missing values
df.dropna(inplace=True)

# Display updated dataset information
df.info()

# Display statistical summary
print(df.describe())
```

```
# Separate features and label
features = df.iloc[:, [0]].values
label = df.iloc[:, [1]].values

# Split the dataset into training and testing sets
from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=10
)

# Create and train the Linear Regression model
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(x_train, y_train)

# Evaluate the model
print("Training Score:", model.score(x_train, y_train))
print("Testing Score:", model.score(x_test, y_test))

# Display coefficient and intercept
print("Coefficient:", model.coef_)
print("Intercept:", model.intercept_)
```

```
# Save the model
import pickle

pickle.dump(model, open('MarksPred.model', 'wb'))

# Load the saved model
model = pickle.load(open('MarksPred.model', 'rb'))

# Predict marks based on study hours
yr_of_exp = float(input("Enter Study Hours: "))

yr_of_exp_NP = np.array([[yr_of_exp]])
Salary = model.predict(yr_of_exp_NP)

# Display the predicted marks
print(
    "Estimated Marks for {} hours of study is {}".format(
        yr_of_exp, Salary
    )
)
```

OUTPUT:

1. Original Dataset

```
    StudyHours  Marks
0            1     35
1            2     40
2            3     45
3            4     50
4            5     55
5            6     60
6            7     65
7            8     70
8            9     74
9           10     79
10          11     83
11          12     87
12          13     91
13          14     94
14          15     97
```

2. Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   StudyHours  15 non-null     int64
 1   Marks       15 non-null     int64
dtypes: int64(2)
memory usage: 372.0 bytes
```

3. Statistical Summary

```
       StudyHours      Marks
count   15.000000  15.000000
mean     8.000000  68.333333
std      4.472136  20.282528
min      1.000000  35.000000
25%      4.500000  52.500000
50%      8.000000  70.000000
75%     11.500000  85.000000
max     15.000000  97.000000
```

4. Linear Regression Model

```
LinearRegression()
```

5. Training Score

```
0.995281820510348
```

6. Testing Score

```
0.9961244386254701
```

7. Model Coefficient

```
array([[4.51970701]])
```

8. Model Intercept

```
array([32.0523195])
```

9. Final Prediction

```
Enter Study Hours: 4
Estimated Marks for 4.0 hours of study is [[50.13114754]]
```

RESULT:

The Linear Regression model was successfully created and used to predict marks based on the number of study hours.